

ATARS v3.1

Automated Time-Series Analysis and Reporting System

System Integration Report — How 9 Verification Modules Work Together

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<https://github.com/Priyanshu-ux712/ATARS>

Version	Year	Description
v2.0.0	2026	Core pipeline: 16 formal operators, Word report, Isolation Forest, Holt-Winters, Runtime Grounding Verifier (RGV), SHA-256 audit log
v3.1.0	2026	System integration: 9 verification modules working together in one pipeline, 18-step data cleaning, statistical inference, automated PPT, Open-Meteo weather

Abstract

Automated reporting systems that use Large Language Models (LLMs) to generate scientific narratives face a fundamental problem: there is no mechanism to verify that the generated text accurately reflects the underlying data. Existing tools such as FActScore and SelfCheckGPT check text against internet sources or other LLM runs — neither approach is suitable for domain-specific numerical data from sensors.

ATARS v3.1 addresses this by integrating 9 verification modules into a single automated air quality reporting pipeline. Each module catches a different category of error that the others cannot: numerical hallucination (RGV), wrong time periods (TCV), causal overreach (CCD), meteorological context (AEE), quality drift over time (RDD), uncertainty omission (UPT), chemical inconsistency (CVCC), vague language (NSS), and missing benchmarks (BNC). A Master Novelty Score (MNS) aggregates all nine into a single auditable trust metric.

The contribution is not any one module individually — it is the integration of all nine into a single pipeline where they work together, each compensating for the blind spots of the others, producing a report that is more trustworthy than any single verification method could achieve alone.

1. The Problem — Why One Verifier Is Not Enough

When an LLM generates a scientific paragraph about air quality data, it can fail in at least 9 distinct ways. Each failure mode is invisible to a verifier designed to catch a different one.

Failure Mode	Example LLM Error	Invisible To
Wrong number	"PM10 annual mean was 83.4 $\mu\text{g}/\text{m}^3$ " when J says 91.2	TCV, CCD, NSS — they don't check numbers
Wrong time period	"PM10 peaked in July" when data shows January peak	RGV, CVCC, NSS — they don't parse temporal claims
False causation	"High humidity causes PM10 to rise"	RGV, TCV, UPT — they don't check causal language
Vague language	"Air quality was poor during winter months"	RGV — no numbers to verify; TCV — no specific claim
Missing uncertainty	"PM10 mean = 91.2 $\mu\text{g}/\text{m}^3$ " with no CI range stated	RGV — number is correct; NSS — sentence is specific
Chemical inconsistency	"NOx was lower than NO2" (chemically impossible)	RGV, TCV, CCD — they don't check variable relationships
Missing benchmark	"91.2 $\mu\text{g}/\text{m}^3$ " with no comparison to WHO or national average	All others — number is correct and specific
Quality drift	G_rate was 92% last run, now 71% — silent degradation	All per-run verifiers — they only see the current run
Wrong meteorological context	Anomaly on 15 Jan not explained by known inversion conditions	All text verifiers — they check prose, not event context

This table shows why a single verifier is insufficient. RGV catches wrong numbers but misses causation errors. CCD catches causation but misses chemical inconsistency. CVCC catches chemical inconsistency but misses temporal claims. Only by running all 9 together does the system catch all 9 failure modes.

2. The System — How 9 Modules Work Together

ATARS v3.1 runs all 9 modules in sequence on every pipeline run. Each module receives the same inputs: the JSON contract J (computed facts), the LLM-generated narrative N, and the daily_df dataset. Each produces a structured result that feeds into the Master Novelty Score.

2.1 The Shared Input: JSON Contract J

J is the formal boundary between computation and language. It contains every number the LLM is permitted to use: annual means, exceedance counts, quality scores, correlation coefficients, anomaly days, and threshold values. J is computed deterministically from the data — it never contains LLM output. All 9 verification modules use J as their ground truth.

2.2 Pipeline Flow

Stage	What Happens	Output Fed To
Data ingestion + 18-step cleaning	Raw CSV → clean dataset with quality score	daily_df, cleaning_report → all modules
Statistical computation (16 operators)	daily_df → J (the ground truth contract)	J → all 9 verification modules + LLM
LLM narrative generation (optional)	J → narrative N (if Ollama running, else placeholder)	N → all 9 verification modules
N-RGV: Numerical grounding	Every number in N checked against J ($\pm 3\%$ tolerance)	G_rate_v2, G_rate_v3 → MNS
N-01 TCV: Temporal claims	Month/season claims in N checked against computed peak/low months	TCV_rate → MNS
N-02 CCD: Causal language	Sentences scanned for causal triggers; flagged if only correlation exists	CCD_score → MNS
N-03 AEE: Anomaly context	Each anomaly day explained by meteorological rules (not LLM)	Explanation table → Section 16b
N-04 RDD: Drift detection	Current G_rate compared against previous audit logs	Drift alert → Section 19
N-05 UPT: Uncertainty	95% CI computed for every J value; claims annotated	CI table → Section 16b
N-06 CVCC: Chemistry	$\text{NO}_x \geq \text{NO}_2 + \text{NO}$ verified in J data	CVCC_score → MNS
N-07 NSS: Specificity	Each sentence scored 0.0–1.0 for specificity	Specificity → MNS (confidence component)
N-08 BNC: Benchmark	Station vs WHO, NAAQS, India national average	Severity → Section 16b
N-09 MNS: Aggregation	Weighted combination of all module outputs → single trust score	MNS → cover page, Section 17, PPT closing
Report assembly	All module results → Section 16b in Word report	Final Word + PPT output

3. Why the Integration Is Novel

Each individual technique used in ATARS has precedent in research literature. The novelty is in the integration — nine different verification methods working together inside a single automated domain-specific reporting pipeline, each compensating for the blind spots of the others.

Component	Precedent in Literature	What ATARS Adds
Numerical grounding	FActScore (Min et al. 2023) — checks text against Wikipedia	Checks against computed sensor data in J, not internet; type-adaptive tolerance
Causal claim detection	Causal NLP papers (Khetan et al. 2022) — detect causal language in text	Integrated into report loop; flags claims where J contains only correlation
Temporal verification	Temporal IE systems — extract time expressions	Verifies claims against computed domain facts (peak month from daily_df)
Anomaly explanation	Rule-based expert systems in meteorology	Automated per-anomaly explanation written into the report output
Benchmark comparison	WHO/CPCB dashboards — static tables	Computed automatically per run and embedded in the report
Quality drift tracking	ML monitoring systems (Alibi-Detect etc.)	Applied to AI narrative quality across report runs, not model predictions
Unified trust metric	Multi-criteria decision making — weighted scoring	Applied to LLM report trustworthiness in a domain-specific pipeline

The key distinction from FActScore and SelfCheckGPT (the closest related work):

- FActScore: checks text against internet sources → not applicable to sensor data
- SelfCheckGPT: checks consistency across multiple LLM runs → requires LLM; treats numbers like any other text
- ATARS RGV: checks numbers against a computed JSON contract → works offline; type-adaptive (concentration $\neq$ count $\neq$ quality score)
- ATARS system: 8 additional modules catch what RGV alone cannot — together they cover all 9 failure modes identified in Section 1

4. Module-by-Module: Role in the System

RGV — Runtime Grounding Verifier (Foundation)

RGV is the foundation layer. It extracts every number from every LLM sentence and checks it against J using type-adaptive tolerance: $\pm 3\%$ for concentrations, ± 2 for integer counts, ± 0.05 for quality scores. A sentence is GROUNDED if all its numbers match J. $G_rate = \text{grounded sentences} / \text{numerical sentences}$.

What RGV cannot catch: a sentence with correct numbers but wrong time period, causal language, vague phrasing, or missing benchmark. This is why the other 8 modules exist.

N-01 TCV — Fills the Temporal Blind Spot

RGV verifies numbers but not when. 'PM10 peaked in July' is not a number — RGV passes it. TCV extracts month and season claims, computes the actual peak and low months from daily_df, and flags mismatches. Tolerance: ± 1 month to account for borderline cases.

N-02 CCD — Fills the Causation Blind Spot

RGV verifies numbers but not their framing. 'Humidity causes PM10 to rise' uses the correct humidity value from J but asserts causation. J only contains Pearson r — correlation, not causation. CCD scans for 26 causal trigger words and flags sentences where J has no causal data to support them. Enforces: $\text{Corr}(X,Y) \neq C(X \rightarrow Y)$.

N-03 AEE — Fills the Context Blind Spot

RGV, TCV, and CCD all operate on LLM text. AEE operates on data. For every anomaly day detected by the z-score or Isolation Forest, AEE applies 8 meteorological rules (wind speed, humidity, temperature, season, day-of-week) to produce a deterministic explanation. This explanation appears in the report regardless of what the LLM wrote. Zero LLM involvement — fully reproducible.

N-04 RDD — Fills the Time-Series Blind Spot

All other modules check a single run. RDD compares the current run against previous audit logs to detect if G_rate , $Q(D)$, or pollutant statistics are drifting. A system that was 92% grounded last week and is 71% this week has a problem — RDD raises the alert.

N-05 UPT — Fills the Uncertainty Blind Spot

RGV checks that a number matches J. But J contains point estimates — mean values without confidence intervals. UPT computes 95% CI for every key statistic and annotates LLM claims with uncertainty bounds. 'PM10 = 91.2 $\mu\text{g}/\text{m}^3$ (95% CI: 89.1–93.3)' is more honest than '91.2 $\mu\text{g}/\text{m}^3$ ' alone.

N-06 CVCC — Fills the Chemical Consistency Blind Spot

RGV checks individual numbers in isolation. CVCC checks relationships between numbers. $\text{NO}_x = \text{NO} + \text{NO}_2$ by chemical definition, so NO_x must always be $\geq \text{NO}_2$. If J contains $\text{NO}_x = 45$ and $\text{NO}_2 = 52$, this is a data quality problem that RGV would miss entirely. CVCC catches it and flags it in the report.

N-07 NSS — Fills the Vagueness Blind Spot

A sentence like 'Air quality was poor during winter' passes RGV (no numbers to check), passes TCV (no specific month claimed), passes CCD (no causal language). NSS catches it: specificity score = 0.0. NSS scores each sentence 0.0–1.0, rewarding numbers, units, formal notation, and WHO references. Vague sentences are flagged.

N-08 BNC — Fills the Context Blind Spot

RGV verifies numbers against J. But J contains station-level data only. BNC compares station results against WHO AQG, CPCB NAAQS, and India national averages — stored internally, no API. 'PM10 = 91.2 $\mu\text{g}/\text{m}^3$ ' is correct per J but meaningless without knowing it is 2.0× the WHO annual guideline and above the India national average. BNC provides that context automatically.

N-09 MNS — The Aggregation Layer

MNS collects one output from each module and combines them into a single weighted score. This is the final answer to: 'How much should I trust this report?' $\text{CERTIFIED} \geq 0.90 \cdot \text{TRUSTED} \geq 0.80 \cdot \text{ACCEPTABLE} \geq 0.70 \cdot \text{REVIEW} \geq 0.55$.

Formula: $\text{MNS} = \text{G_v2} \times 0.25 + \text{G_v3} \times 0.20 + \text{TCV} \times 0.15 + \text{CCD} \times 0.15 + \text{CVCC} \times 0.10 + \text{Conf} \times 0.10 + \text{Sem} \times 0.05$

5. What the System Catches — A Complete Example

Suppose the LLM generates this sentence: 'High humidity in July caused PM10 to peak at 83.4 $\mu\text{g}/\text{m}^3$, significantly above normal levels.'

This single sentence has four errors — caught by four different modules:

Error in Sentence	Module That Catches It	How
'83.4 $\mu\text{g}/\text{m}^3$ ' — wrong value (J says 91.2)	RGV	91.2 vs 83.4: difference = 8.6% > $\pm 3\%$ tolerance $\rightarrow$ UNGROUNDED
'in July' — wrong month (data peaks in January)	TCV	Claimed month = July, actual peak month = January $\rightarrow$ INCORRECT
'caused' — causal language, only correlation in J	CCD	Trigger word 'caused' found; J has only Pearson r $\rightarrow$ FLAGGED
'significantly above normal' — no number, vague	NSS	No numbers, no units, no formal notation $\rightarrow$ specificity = 0.0

No single module catches all four errors. RGV misses the month. TCV misses the wrong number. CCD misses the wrong number. NSS misses the causation. Together they catch everything.

6. V2 G_rate = 100% vs V3 G_rate = 94.4% — Explained

When `--no-llm` is active, ATARS uses statistical placeholders built directly from J values. Verifying placeholder text against the same J that created it is circular — both scores are not meaningful in this mode. Real verification only applies when Ollama generates the narrative.

	V2 RGV	V3 Enhanced RGV
Tolerance	±5% for all numbers	±3% for concentrations, ±2 for counts, ±0.05 for scores
Sources	J contract only	J + WHO AQG + live weather values
Semantic check	No	Yes — checks meaning, not just numbers
Contradiction detect	No	Yes — flags direction words vs actual data direction
Result (placeholder mode)	100% (circular)	94.4% (tighter tolerance catches more mismatches)
Result (real LLM mode)	Meaningful measure	Stricter meaningful measure

94.4% with `--no-llm` is correct behaviour: V3 is a genuinely harder test. With a real LLM, both scores become meaningful — V3 will always be lower than or equal to V2 because its standards are higher.

7. Other New Features in v3.1

7.1 18-Step Data Cleaning Engine

Professional data analyst-grade cleaning before any statistical computation. 18 sequential steps including flatline detection, rate-of-change spike filtering, $3\times$ IQR Tukey winsorization, 5-strategy imputation cascade, chemical constraint audit, and a 7-dimensional quality score producing EXCELLENT/GOOD/FAIR/POOR verdict. Every step is logged and shown in Section 00b of the Word report.

7.2 Formal Statistical Inference Layer

Six inferential statistics modules beyond v2.0 descriptive stats: Mann-Kendall trend test (non-parametric, $O(n \log n)$), normality tests (Shapiro-Wilk + D'Agostino), OLS with full inference (SE, t-stats, VIF, Durbin-Watson), seasonal decomposition (Trend + Seasonal + Residual), exceedance frequency analysis (return periods, excess burden). All pure Python/numpy — no statsmodels dependency.

7.3 Open-Meteo Live Weather (18 Variables)

Free ERA5 reanalysis + forecast API. No API key. Batch-fetches all dates in one call. SSL certificate verification disabled for Windows compatibility. 18 raw variables (temperature, wind, precipitation, radiation, humidity, pressure, dewpoint, weather code) + 18 derived indices (inversion risk, fog risk, dust risk, stagnant air, pollution risk score 0–6).

7.4 Automated 19-Slide PPT Generator

`generate_summary_ppt()` produces a complete executive presentation with every run. Slide 4 uses a two-column layout: Q(D) score on the left, per-variable bars on the right. Closing slide (slide 15) shows four live KPI stat cards (Days / Q(D) / G_rate / MNS) computed from the current run — not hardcoded values.

8. References

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